Can a statistics machine pass the German state examination?

Replicating LEXam on German First State Exam-style essay cases

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1,127 German exam cases · 14 models under the LEXam protocol · 37 human participants · ~23,000 newly graded answers

Research questions

Language models are statistics machines: never taught legal method, only trained to predict likely text. Yet their answers look like proper Gutachten - Obersatz, definition, subsumption, conclusion.

- **RQ1:** How well do they actually do on German state-exam-style cases, measured with an established method (LEXam)?
- **RQ2:** Does that established method fit German correction practice - and what changes when a German doctrinal rubric does the grading instead?
- **RQ3:** How do the machines compare with humans who wrote the same exams?

Behind all three: if a machine that only learned patterns scores well, then part of what our exams reward is itself a learnable pattern. The answer tells us something about the machines *and* about the exams.

What we did

- **Replicated LEXam** (Fan et al. 2026, University of Zurich) faithfully on the German BenGER corpus: generation prompt and judge prompt **verbatim**, only Swiss law substituted for German law. Zero-shot - no system prompt, no examples, no retrieval.
- The LEXam judge is an LLM examiner that scores each answer **against the reference solution** (0-100).
- Deviations (engineering, not method): one judge model instead of the 3-judge ensemble, structured JSON output, MCQ track adapted to yes/no.
- 0 judge errors over ~23,000 cells. Total cost under €100. Prompts, configs, and analysis code published.

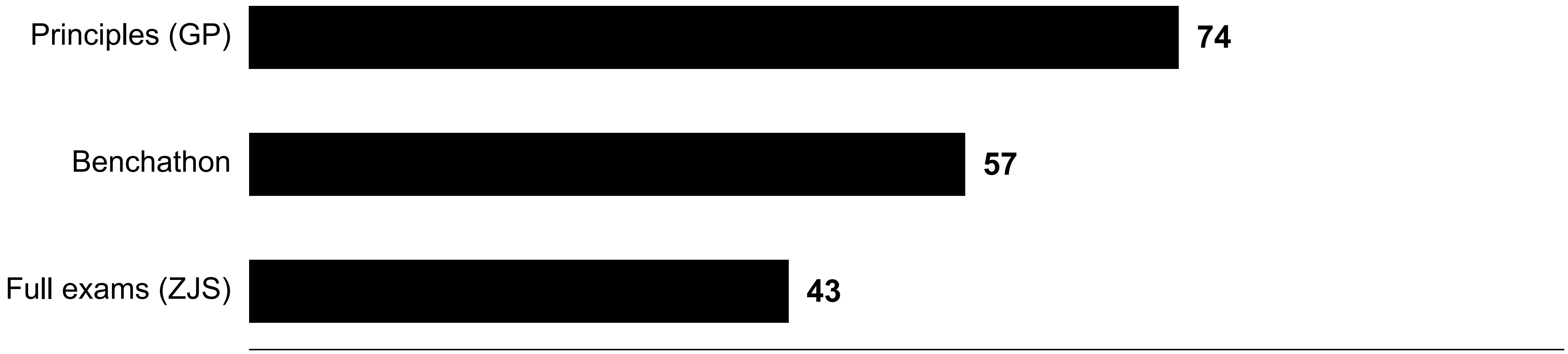
Corpus and models

Tier	Cases	What it is
Grundprinzipien	531	short yes/no principle questions (civil & criminal law)
Benchathon	15	curated exam cases, also answered by 37 humans (220 solutions)
ZJS Fälle	581	full practice exams in state-exam format; reference solutions avg. 30,000 characters

- **14 affordable-tier models** from every major provider, incl. the 3 with published Swiss LEXam scores.
- **Flagships** (GPT-5.4, Claude Opus, Gemini Pro) join the German-rubric analysis.

The replication result

Best model per tier, LEXam correctness judge (0-100). Competent on principle questions, overwhelmed by the full exam:



- The three models with published Swiss LEXam scores land **15-22 points lower** on German full exams and 5-11 higher on principle questions - the multi-hour Gutachten format is the hurdle, not the language.

Why the LEXam judge does not fit the German Klausur

- The LEXam judge anchors every score to **one reference solution**. That is its strength - a strict, reproducible measure of substance.
- But German correction accepts **vertretbar alternative paths**: an answer may deviate from the Musterlösung and still be doctrinally sound. A single-reference anchor **penalises exactly these answers** - ~6.5% of ZJS answers sit in this spot.
- And German grading practice officially rewards **craft** - Gliederung, method, language, ~30 of 100 points. A correctness-vs-reference score rightly excludes craft; a benchmark for German legal education must include it.
- **So we graded everything a second time ourselves:** a 100-point German doctrinal rubric (10 dimensions of German correction practice, mapped to Notenpunkte) - same cases, each pipeline with its own prompt and its own judge.

Two graders, same 581 Klausuren - the numbers

Model	LEXam judge	German rubric	Δ
GLM-5.2	43.3	49.8	+6.5
gpt-5.4-mini	39.9	49.7	+9.8
DeepSeek-V4-Flash	39.2	45.9	+6.7
Gemini-3.1-Flash-Lite	39.2	44.6	+5.4
Qwen3.5-122B	36.6	44.2	+7.6
Qwen3.6-27B	36.0	42.5	+6.5
Qwen3.6-35B-A3B	33.3	39.7	+6.4
Llama-4-Maverick	32.3	36.1	+3.8
Gemma-4-26B	32.3	35.5	+3.2
custom (anonymised)	26.6	28.9	+2.3
GLM-4.7-Flash	24.4	25.5	+1.1

Per-model mean, 0-100, ZJS full exams, one (latest) generation per task and model. The 11 models covered by both pipelines: 3 affordable models were never graded under the rubric, flagships never under the LEXam judge. Δ = rubric - judge.

What the comparison shows

- **The rankings agree exactly:** across all 11 models, the correctness judge and the German rubric produce the identical ordering. LEXam's lean single-reference recipe survives the jurisdiction change and a much longer exam format - and our rubric's substance core moves with the established method.
- **The levels differ:** the rubric grades 1-10 points higher, and the gap grows with model strength - consistent with fluent systems collecting most of the ~30 craft points the correctness judge ignores.
- **What a German corrector would write:** under the rubric, flagships pass 76-83% of full Klausuren (6.5-7.3 grade points, a solid "befriedigend"); real student cohorts average 2-6 grade points.
- Neither number replaces the other: the judge shows what an answer is worth with form stripped away; the rubric shows the German grade under it.

Humans vs. machines - under both graders

Benchathon cases, same exams for humans and models:

Group	LEXam judge (0-100)	German rubric (0-100)
Affordable-model field average	42	not run
Unaided participants	47	50
Best affordable model	57	not run
Human-AI co-creation	62	66
Flagships	not run	67-69

The second agreement: the co-creation uplift measures ~15 points under both instruments (+14.9 judge, +15.7 rubric). Two independently built graders detect the same effect.

"not run": that pipeline was not executed for this group - the affordable replication models were not graded under the rubric on Benchathon, and the flagships were not run under the LEXam judge.

Takeaways

- **Legal education:** flawless structure is now cheap - for every student and every machine. The discriminating signal has moved to issue-spotting, weighting, and the decisive doctrinal step. Exam design should follow it there.
- **The profession:** part of legal argument is demonstrably pattern-like; machines master it. The rest is identifiable, teachable method - and remains human territory for now.
- **Benchmarking:** the field needs a common validated foundation (LEXam) plus jurisdiction-specific instruments on top (BenGER). Whoever configures the grader defines legal quality.

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